

Week 6

Debt

Bills & Valuation

Bonds & Valuation

This Week We will:

6.1: Overview of Australian Financial Market -SLO2

6.2: Overview of Debt -SLO2

6.3: Commercial bill & Valuation -SLO2

6.4: Bond & Valuation -SLO2

6.4 a: How to use a Financial Calculator to calculate Bond Price

Learning Objectives:

At the end of this module you will:

SLO2: apply time-value of money techniques to the valuation of financial instruments and financial securities

Overview of Australian Financial Market

- *Australian Financial markets*
- A place that:
- **Provides funds** to businesses (e.g: businesses issuing debt or equity securities to raise funds)
- **Provides a value** for those securities (what is the market value for a loan or share in today's value) – **Price discovery**
- Provides a mechanism for **transferring the ownership** of those securities (e.g: investors can provide funds to business in return for shares in the business)

Australian Financial Markets

- Many financial markets:
 - Foreign exchange
 - Futures and options
 - Debt markets
 - Equity markets

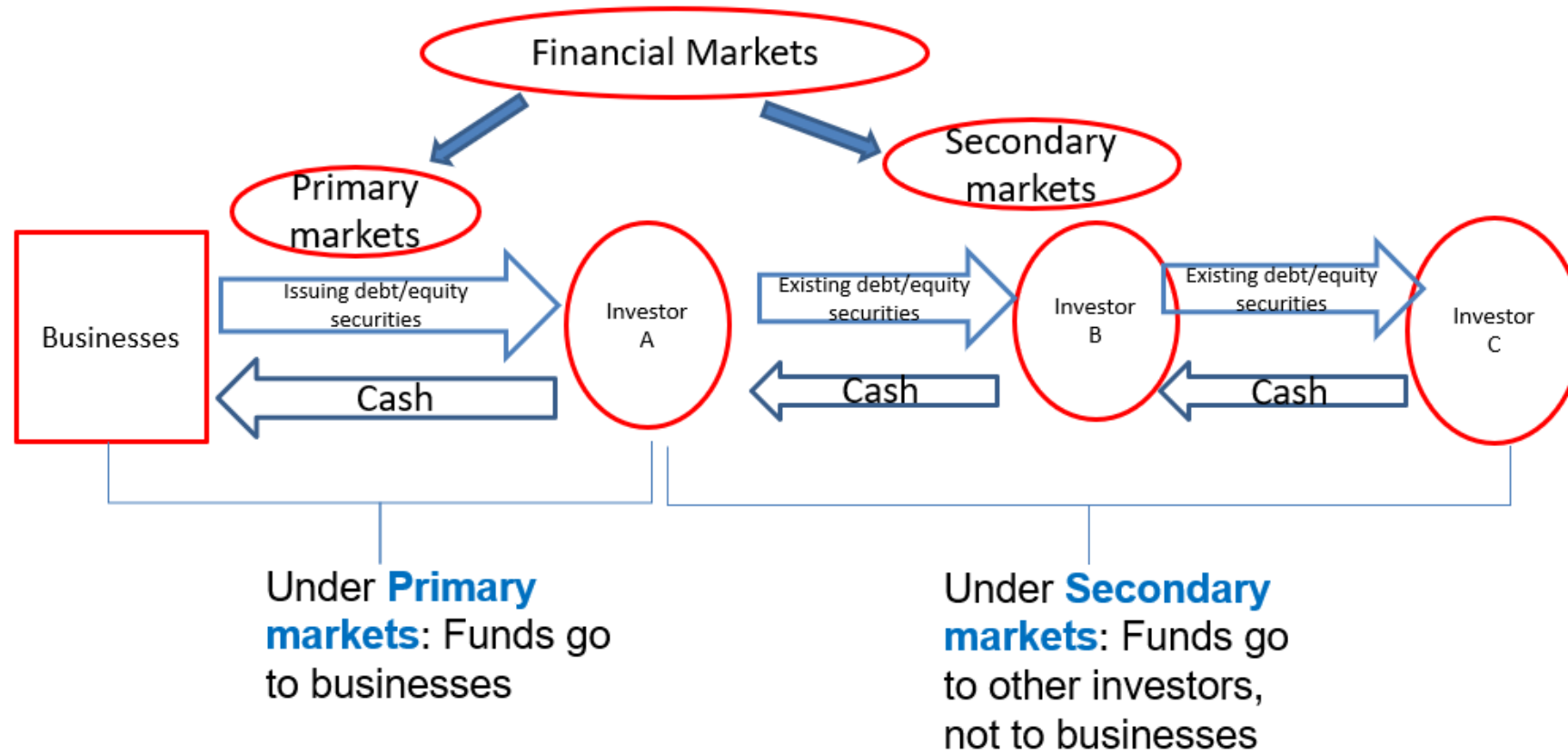


Capital markets:

**Our main
focuses in this
subject**

Primary markets vs Secondary markets

- Australian Financial Markets function as both **primary markets** and **secondary markets**
- **Primary markets** are where firms raise capital by issuing their securities for the first time (e.g: Initial Public Offering-IPO)
- **Secondary market** are where ownership of securities is transferred from one investor to another (e.g: Buying shares from Australian Stock Exchange-ASX)



Question 1

Jean is having issues in distinguishing primary market and secondary market through the following transactions.

As a finance manager, please advise Jean on the following transactions:

1. You buy Apple shares through a stock exchange
2. Start-up “XYZ” issues their shares to the public to raise funds for their upcoming projects
3. Your friend purchased the “XYZ” share
4. and immediately resells the share to you at the same price as he purchased

Debt versus Equity

- Debt

- Not an ownership interest
- Creditors do not have voting rights
- Interest is considered a cost of doing business and is tax deductible
- Creditors have legal recourse if interest or principal payments are missed
- Excess debt can lead to financial distress and bankruptcy

- Equity

- Ownership interest
- Common stockholders vote for the board of directors and other issues
- Dividends are not considered a cost of doing business and are not tax deductible
- Dividends are not a liability of the firm, and stockholders have no legal recourse if dividends are not paid
- An all-equity firm cannot go bankrupt

Overview of Debt

	Debt
Definition	• An amount of money borrowed today from a lender to be repaid in the future • It's a contractual obligation which commits the borrower to interest and principal payments
Payments	• Interest must be paid. Principal must be repaid. • Default is failure to meet the required payments when due - >Default risk
Tax deductibility	• Interest is a business expense, so it is tax deductible. • Principal payment is not tax deductible
Voting power	• No voting power

Where can you borrow?

- **Financial Institutions**

Banks, credit unions

Provide loans to business and individuals

Often require **security**

Interest rate can be **fixed** or **floating**

Repayments can vary

- Interest Only
- Principal & Interest (P&I)
- Interest at end

Where can you borrow?

- **Markets**

Money or bond market

Business raise funds by issuing securities (**primary**)

Debt securities show that a holder is owed money

Debt securities are tradable (**secondary**)

Interest payments depend on term of the debt issued

Risks Associated with Debt

- **Interest Rate Risk**
 - Changes in interest rates mean borrower's repayments may change
- **Default Risk**
 - The borrower may run out of cash and not be able to make repayments of principal and/or interest
- **Currency Risk**
 - Changes in exchange rates can affect repayments if funds are sourced from overseas markets

How do Lenders Manage Risk?

- **Security**

An agreement for a lender to take possession of the borrower's assets if they default

Not the same meaning as the “security” which is issued

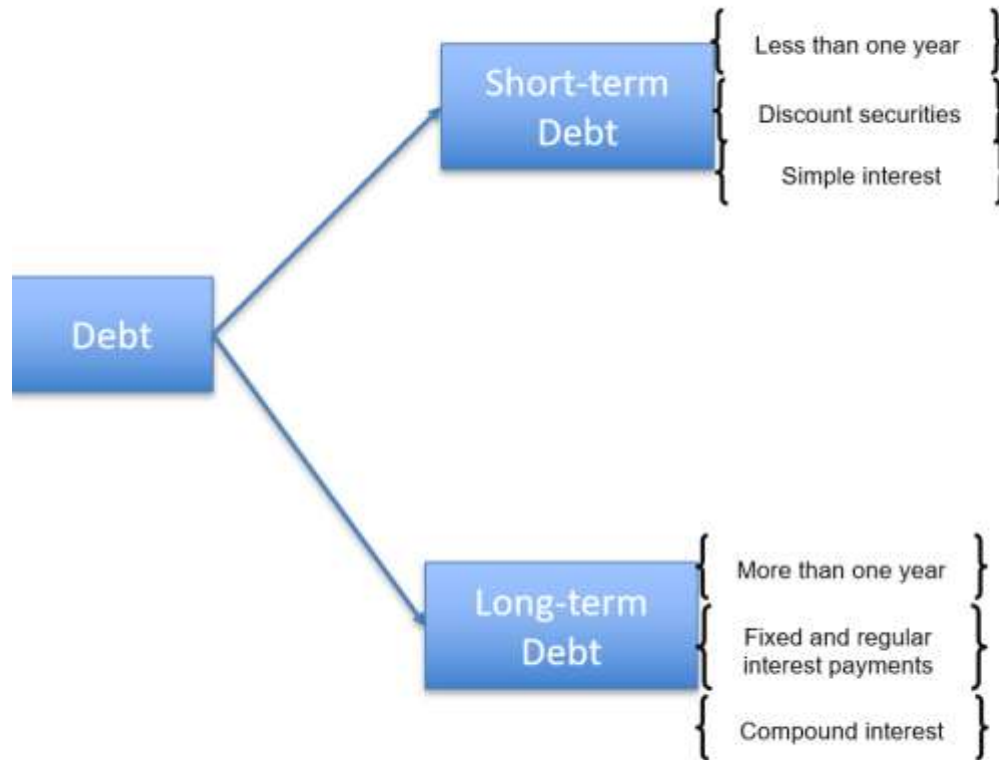
- **Floating charge** over assets
- **Mortgage** over land or building
- **Specific charge** over an asset

- **Loan Agreements**

Specific requirements on the business borrowing funds

- Maintain **certain financial ratios**
- **Cannot sell assets** without consent
- **Guarantees** from a third party

Different Types of debt



Example

COMMERCIAL
BILLS

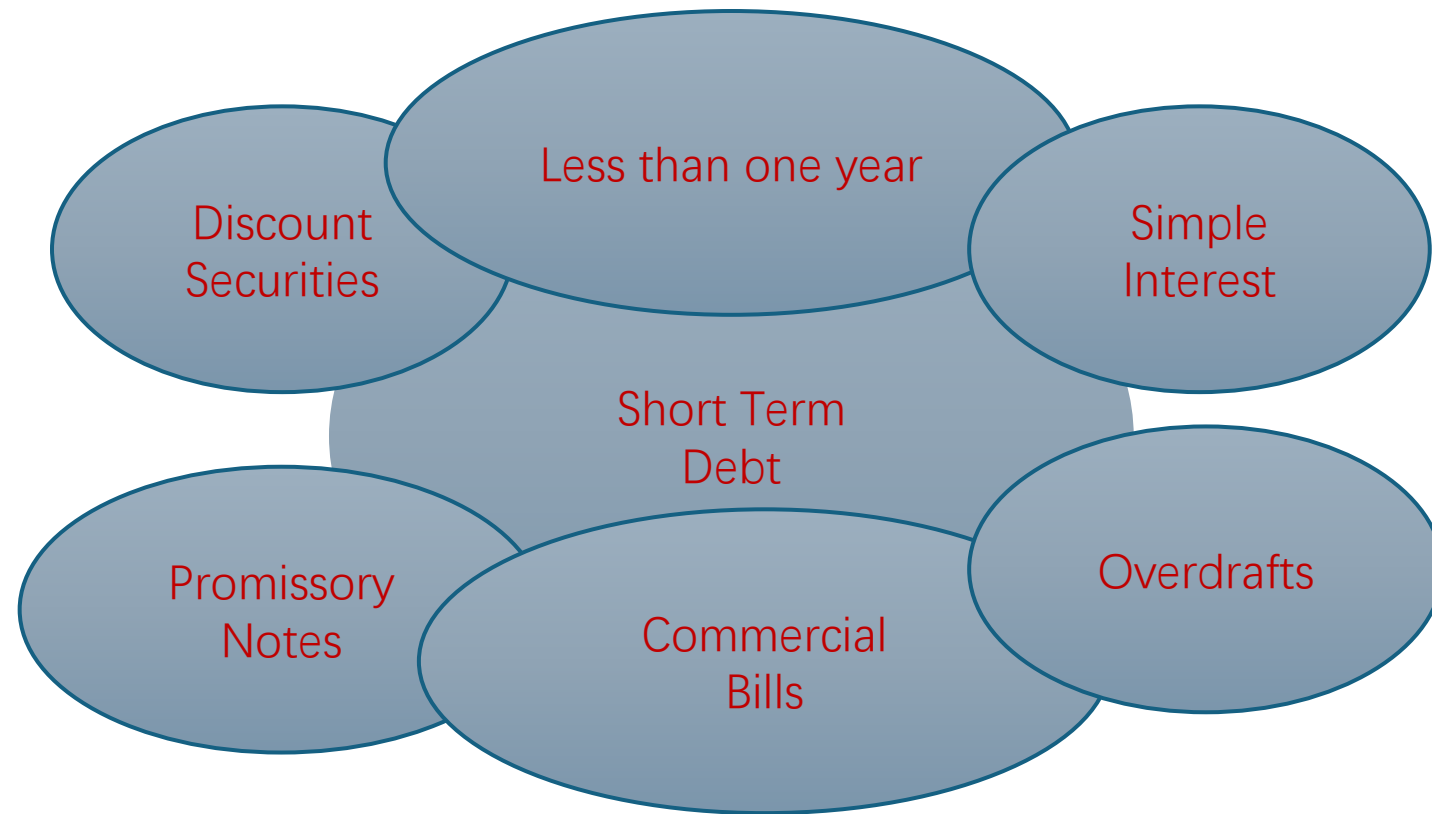
BONDS

Question 2

- Looking for debt options to raise funds for a project is part of:
 - a. Investment decision
 - b. Financing decision
 - c. Working capital decision

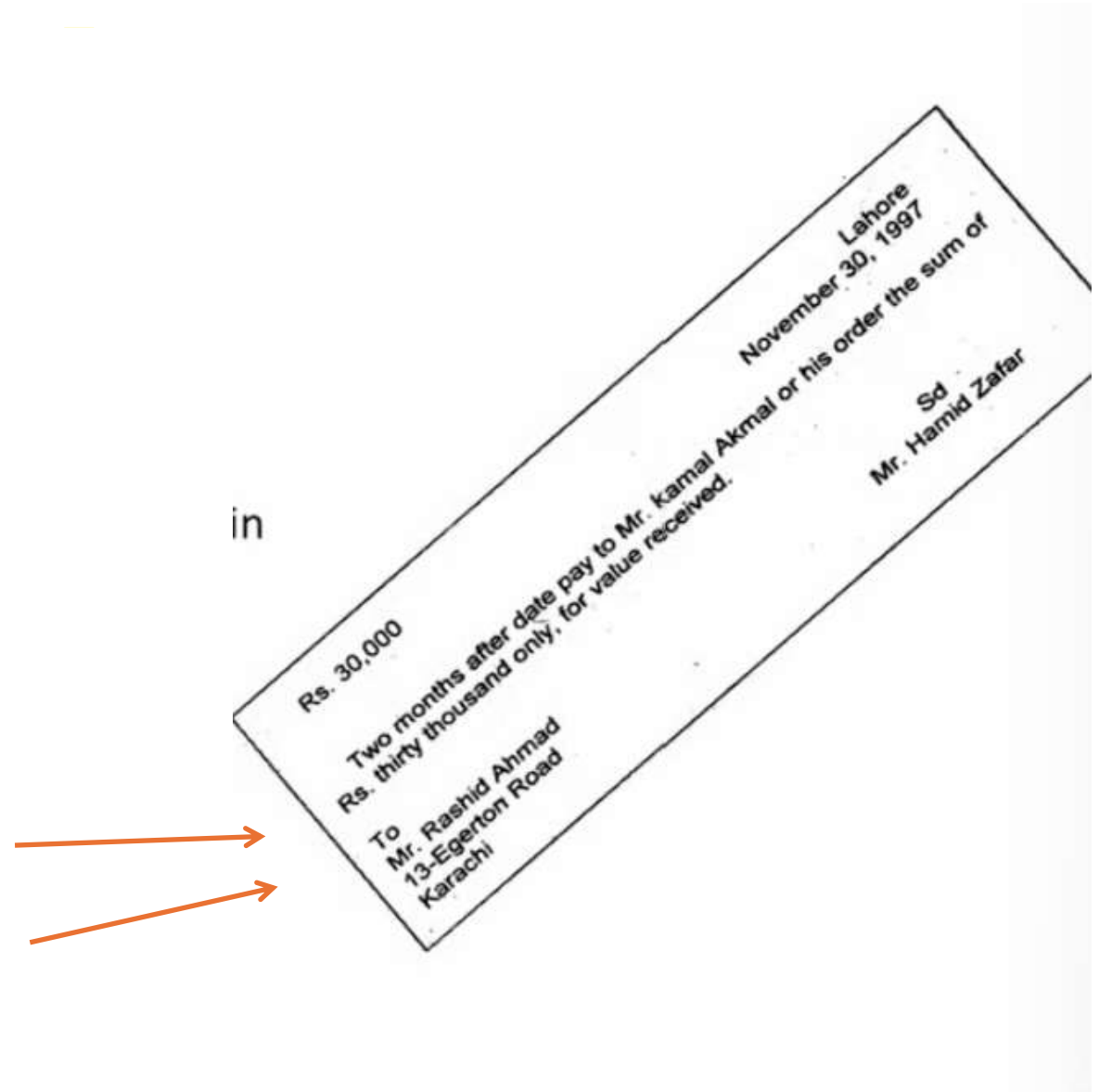
Answer: b

Short Term Debt



Commercial Bills

- **Short term debt security** issued by a business to raise funds in short-term
- It is **unsecured**
- It is a 'promise to pay' at some point in the future
- A bill states its:
 - Face Value**—its future value
 - Maturity Date**—the date it is payable
- Bills are 'discount securities' so we **discount the face value to find the present value** or price of the bill
- Bills are valued using **simple interest**



Parties to a Bill

- The **Borrower** also called the “**drawer**”.

- will **draw** the bill, sign it and **issue it to raise funds**.

- The **Lender** also called the “**discount**er”

- will **buy** the bill at its present value (**discount value**)

- will **receive the Face value**(future value)at maturity.

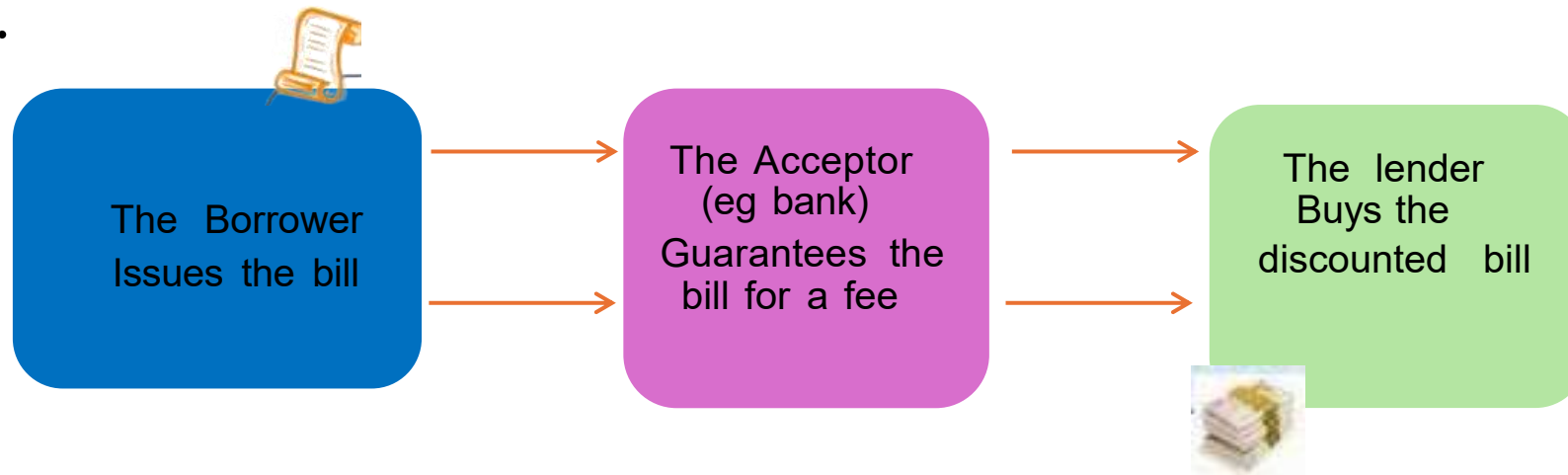
- **Acceptor**—a high creditworthy organisation (eg Bank)
'accepts' the bill and acts as intermediary.

- **guarantees** the payment of the bill at maturity. They pay the bill at maturity then collect from the borrower.

- get a **fee** from the borrower for this guarantee.

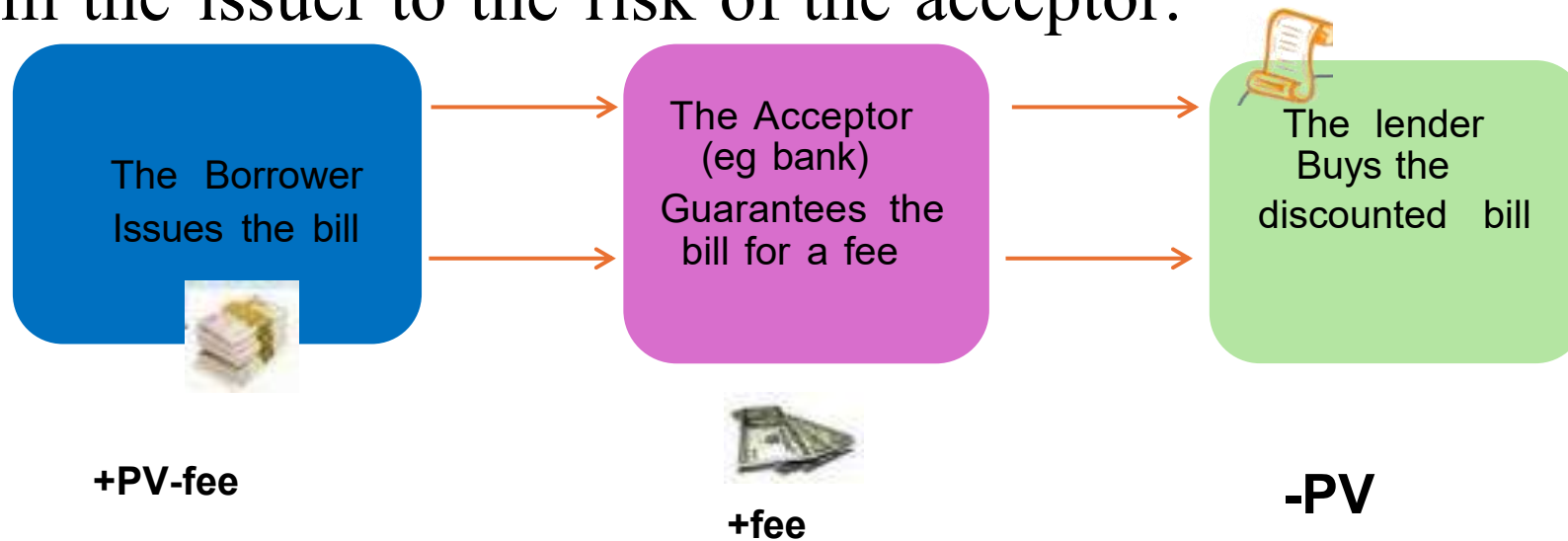
Issuing a Bill

The 'acceptor' acts as intermediary and changes the risk of the bill from the issuer to the risk of the acceptor.



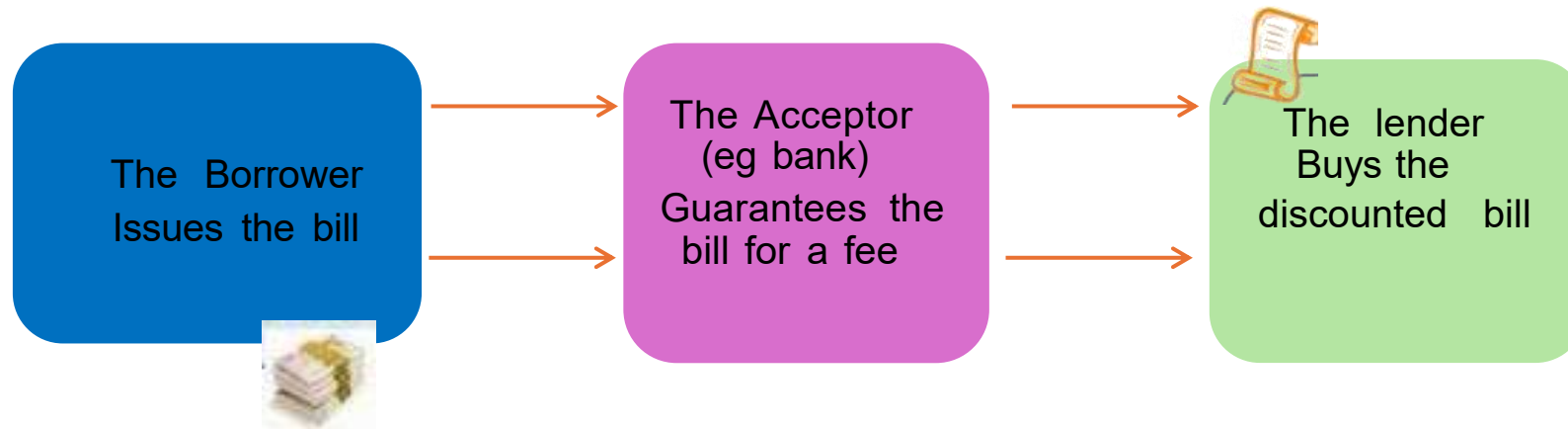
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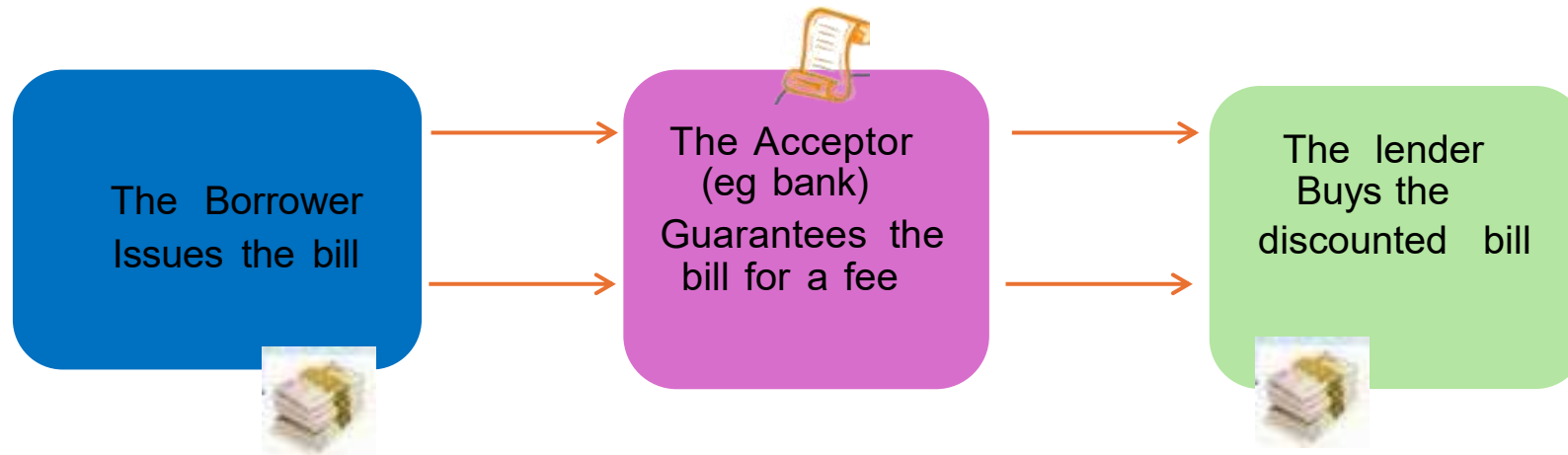
Issuing a Bill

The final investor (discounters) will be paid the FV by Accetor at maturity



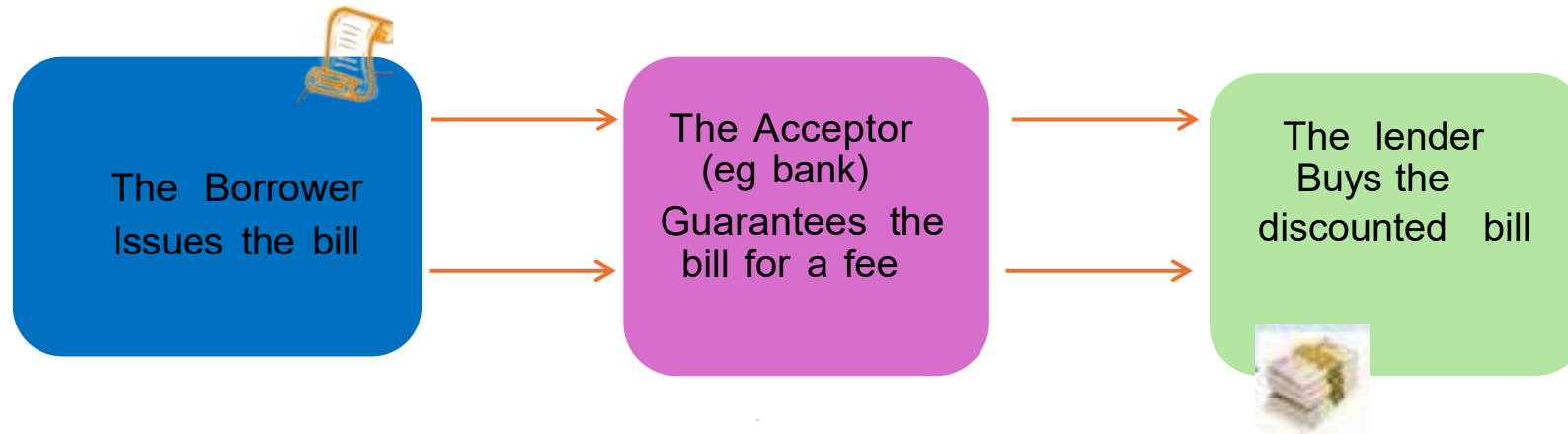
Issuing a Bill

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Issuing a Bill

The final investor (discounters) will be paid the FV by Accetor at maturity



Bill valuation

- Bills are **discount instruments**
- Its value today is the present value of its Face Value calculated using **simple interest**

$$PV = \frac{FV}{(1 + i \times n)}$$

i:interest rate per period

n:time to maturity

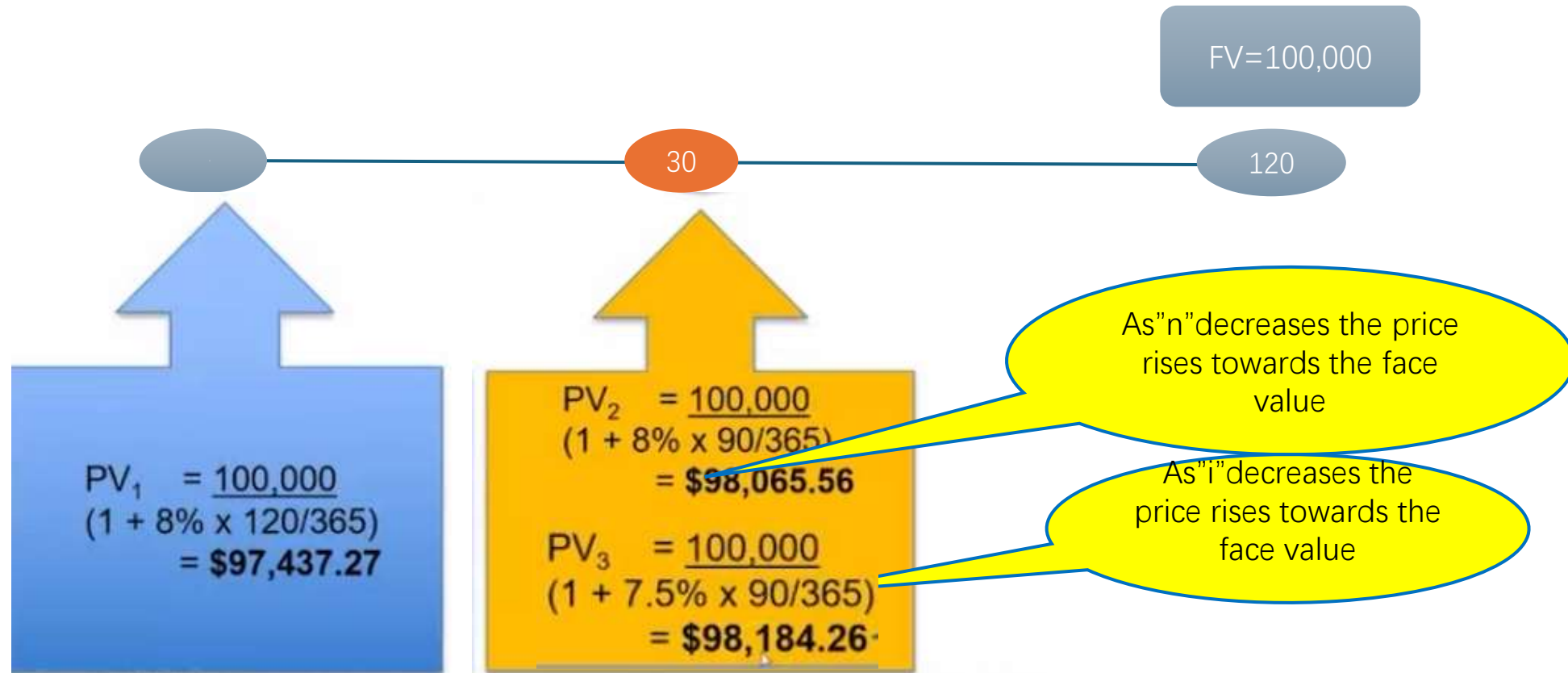
Practice 1

Suppose you purchase a bank accepted bill with a face value of \$100,000. It has 120 days to maturity and is trading at an interest rate of 8% pa.

- 1) What was the purchase price today (Day 0)?
- 2) Suppose you sell the BAB 30 days later. What was the sale price, if the interest rate was 8.0%?
- 3) (Still assuming 30 days later) What was the sale price if the interest rate was 7.5%?

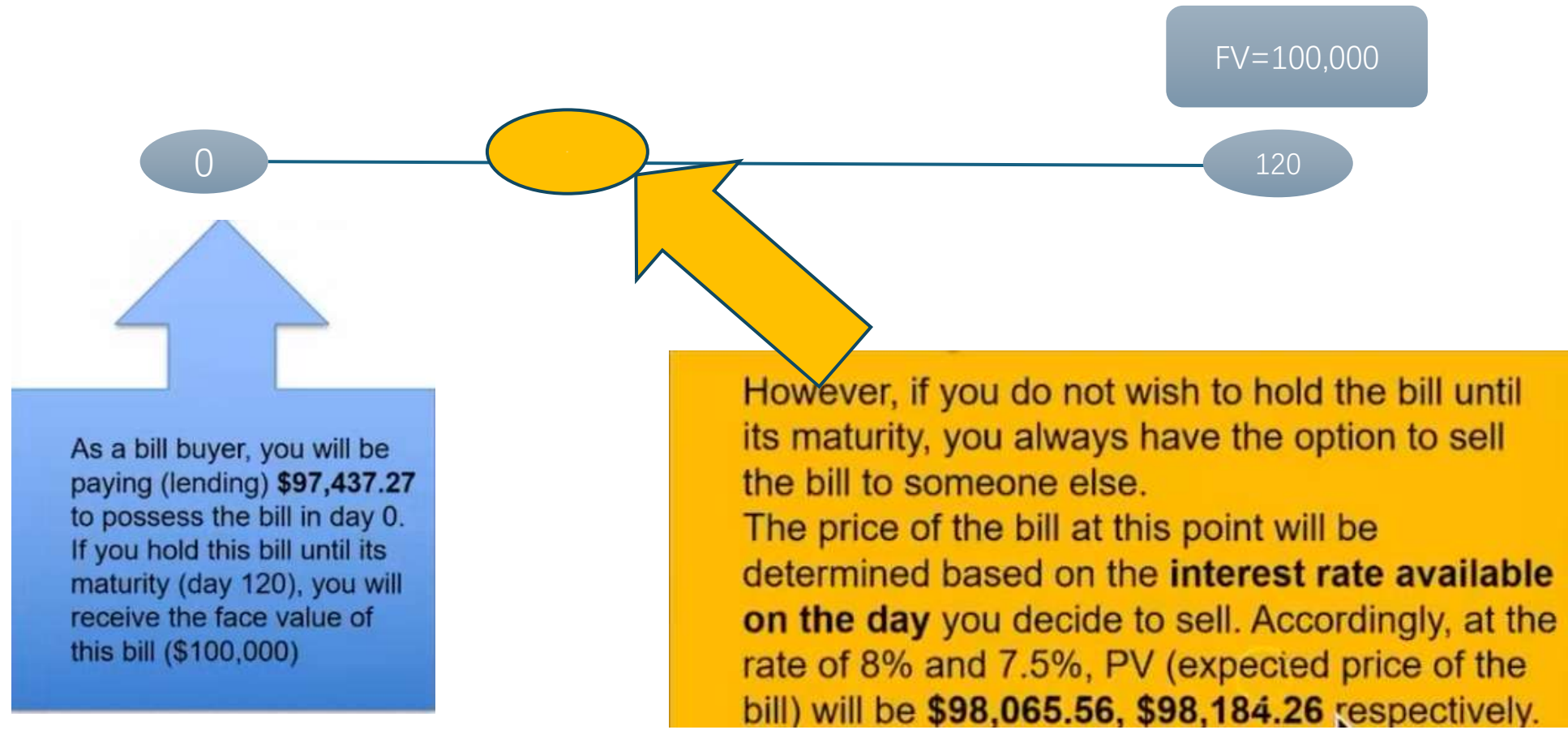
Practice1 Solution

A bank accepted bill with a face value of \$100,000. It has 120 days to maturity and is trading at an interest rate of 8% pa. **(1)** Purchase price today? **(2)** Price 30 days later at 8.0%? **(3)** Price 30 days later at 7.5%?



Practice 1 Solution

A bank accepted bill with a face value of \$100,000. It has 120 days to maturity and is trading at an interest rate of 8% pa. **(1)** Purchase price today? **(2)** Price 30 days later at 8.0%? **(3)** Price 30 days later at 7.5%?



The Bond Indenture

- Contract between the company and the bondholders that includes:
 - The basic terms of the bonds
 - The total amount of bonds issued
 - A description of property used as security, if applicable
 - Sinking fund provisions
 - Call provisions
 - Details of protective covenants

Sample Bond Features

- Features of a recent Pepsico bond issue demonstrate the range of topics covered in the bond indenture:

Sample Bond Features

FEATURES OF A PEPSICO BOND		
TERM		EXPLANATION
Amount of issue	\$1 billion	The company issued \$1 billion worth of bonds.
Date of issue	01/15/2010	The bonds were sold on 01/15/2010.
Maturity	01/15/2040	The bonds mature on 01/15/2040.
Face value	\$2,000	The denomination of the bonds is \$2,000.
Annual coupon	5.50%	Each bondholder will receive \$110 per bond per year (5.50% of face value).
Offer price	98.927	The offer price will be 98.927% of the \$2,000 face value, or \$1,978.54, per bond.
Coupon payment dates	01/15, 7/15	Coupons of $\$110/2 = \55 will be paid on these dates.
Security	None	The bonds are not secured by specific assets.
Sinking fund	None	The bonds have no sinking fund.
Call provision	At any time	The bonds do not have a deferred call.
Call price	Treasury rate plus 0.15%	The bonds have a "make whole" call price.
Rating	Aa2, A+	The bonds are in the middle of the investment grade rating.

Bonds and Bond Valuation

- A bond is a legally binding agreement between a borrower and a lender that specifies the:
 - Par (face) value
 - Coupon rate
 - Coupon payment
 - Maturity Date
- The yield to maturity is the required market interest rate on the bond.
- Do not confuse the coupon rate with the required market interest rate

Practice 2

- A bond has a face value of \$10,000. There are three years to maturity and the coupon rate is 7% pa and coupons are paid semi-annually.

What is the coupon payment?

➤ $PMT =$
 $=$

- What is the number of coupons remaining?

➤ $n =$

Practice 2 Solution

- A bond has a face value of \$10,000. There are three years to maturity and the coupon rate is 7% pa and coupons are paid semi-annually.

What is the coupon payment?

- $$\begin{aligned} \text{PMT} &= 7\% / 2 \times 10,000 \\ &= \$350 \end{aligned}$$

- What is the number of coupons remaining?

- $$n = 3 \times 2 = 6$$

Bond Valuation

- Primary Principle:
 - Value of financial securities = PV of expected future cash flows
- Bond value is, therefore, determined by the present value of the coupon payments plus the present value of the par, or face, value
- Interest rates are inversely related to present (i.e., bond) values.

The Bond-Pricing Equation

$$\text{Bond Value} = C \left[\frac{1 - \frac{1}{(1+r)^T}}{r} \right] + \frac{F}{(1+r)^T}$$

Notice that:

- ❑ The first term is the present value of the coupon payments (an annuity); and,
- ❑ The second term is the present value of the bond's par value

Example 1

- ❑ What is the price of a bond that was issued two years ago and has eight years to maturity?
- Coupons are paid half yearly and a coupon payment was made today.
- The coupon rate is 5% pa and the current market yield is 6% pa.
- The face value is \$200,000

Example 1 Solution

Number of payments per year = 2

FV = \$200,000, coupon rate = 5% pa

PMT = \$200,000 x 0.05/2 = \$5,000

What is the market yield?

$i = 6\%/2 = 3\%$

Years to maturity = 8

number of compounding periods = $2 \times 8 = 16$

Example 1 Solution

$$PV = C \left[\frac{1 - \frac{1}{(1 + r)^T}}{r} \right] + \frac{F}{(1 + r)^T}$$

$$\begin{aligned} PV &= 5000 \left[\frac{1 - (1 + 0.03)^{-16}}{0.03} \right] + 200000(1 + 0.03)^{-16} \\ &= 187,438.90 \end{aligned}$$

Example 2

- Xanth Co. has issued a 10 year bond with an 8% annual coupon. The cash flows from the bond would be paid as follows:

Cash flows

Year	0	1	2	3	4	5	6	7	8	9	10
Coupon		\$80	\$80	\$80	\$80	\$80	\$80	\$80	\$80	\$80	\$ 80
Face value											1,000
		<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$80</u>	<u>\$1,080</u>

As shown, the Xanth bond has an annual coupon of \$80 and a face, or par, value of \$1,000 paid at maturity in 10 years.

Example 2:

- **First**, at the going rate of 8 percent, the present value of the \$1,000 paid in 10 years is:
 - **Present value** = $\$ 1,000 / 1.08^{10}$
= $\$ 1,000 / 2.1589$
= $\$ 463.19$
- **Second**, the bond offers \$80 per year for 10 years; the present value of this annuity stream is:
 - **Annuity present value** = $\$ 80 \times (1 - 1/1.08^{10}) / 0.08$
= $\$ 536.81$

Example 2:

- We can now add the values for the two parts together to get the bond's value:
- **Total bond value = \$ 463.19 + 536.81**
= \$ 1,000

The bond is called **par value bond.**

Example 2:

- **Suppose the YTM rises to 10%, what is the value of the bond?**

First, at the going rate of 10 percent, the present value of the \$1,000 paid in 10 years is:

- **Present value = \$ 1,000 / 1.10¹⁰**
= \$ 1,000 / 2.5937
= \$ 385.55

- **Second**, the bond offers \$80 per year for 10 years; the present value of this annuity stream is:

- **Annuity present value = \$ 80 × (1 - 1 / 1.10¹⁰) / 0.10**
= \$ 491.57

Example 2:

- We can now add the values for the two parts together to get the bond's value:
- **Total bond value = \$ 385.55 + 491.57**
= \$ 877.12
- The bond is called **discount bond**.

Example 2:

- Suppose the YTM falls to 6%, what is the value of the bond?

First, at the going rate of 6 percent, the present value of the \$1,000 paid in 10 years is:

- **Present value** $= \$1,000 / 1.06^{10}$
 $= \$1,000 / 1.7908$
 $= \$558.41$

- Second, the bond offers \$80 per year for 10 years; the present value of this annuity stream is:

- **Annuity present value** $= \$80 \times (1 - 1/1.06^{10}) / 0.06$
 $= \$588.81$

Example 2:

- We can now add the values for the two parts together to get the bond's value:
- **Total bond value = \$ 558.41 + 588.81**
= \$ 1147.22
- The bond is called **premium bond**.

Question 3

A premium bond has a coupon rate that:

- A) is less than the yield to maturity.
- B) equals zero.
- C) must be variable
- D) equals the current yield
- E) exceeds the yield to maturity

The answer is: E

Question 4

❑ A bond has a \$1,000 face value, a market price of \$980, and coupon payments of \$35 every six months. What is the annual coupon rate?

A) 3.50 percent

B) 7.00 percent

C) 6.86 percent

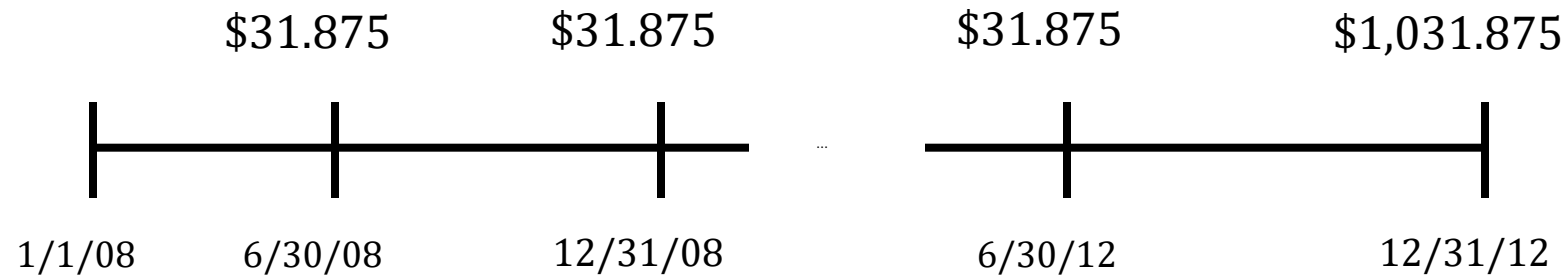
D) 3.57 percent

E) 7.14 percent

The answer is B

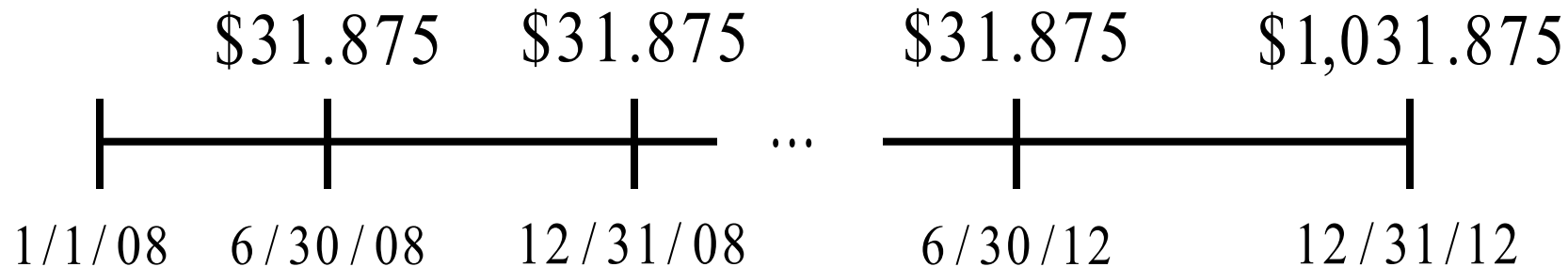
Example 3

- Consider a U.S. government bond with a $6\frac{3}{8}\%$ coupon that expires in December 2012.
 - The *Par Value* of the bond is \$1,000.
 - Coupon payments* are made **semi-annually** (June 30 and December 31 for this particular bond).
 - Since the *coupon rate* is $6\frac{3}{8}\%$, the payment is \$31.875.
 - On January 1, 2008 the size and timing of cash flows are:



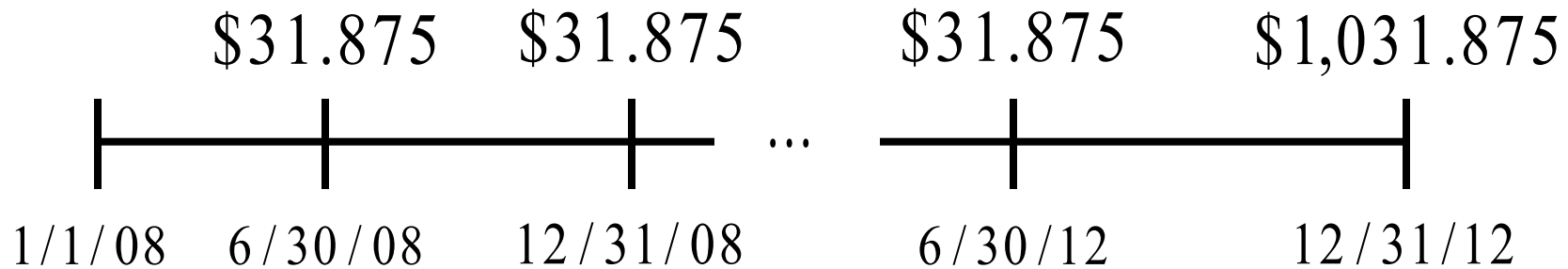
Example 3:

- Consider a U.S. government bond with as 6.375% coupon that expires in December 2012.
 - The *Par Value* of the bond is \$1,000.
 - Coupon payments* are made **semi-annually** (June 30 and December 31 for this particular bond).
 - Since the *coupon rate* is 6.375%, the payment is \$31.875.
 - On January 1, 2008 the size and timing of cash flows are:



Example 3:

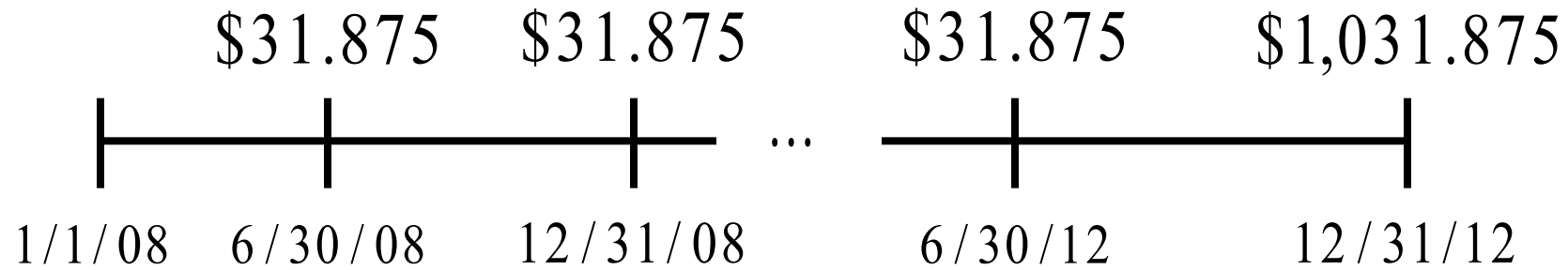
- On January 1, 2008, the required yield is 5%.
- The size and timing of the cash flows are:



$$PV = \frac{\$31.875}{.05/2} \left[1 - \frac{1}{(1.025)^{10}} \right] + \frac{\$1,000}{(1.025)^{10}} = \$1,060.17$$

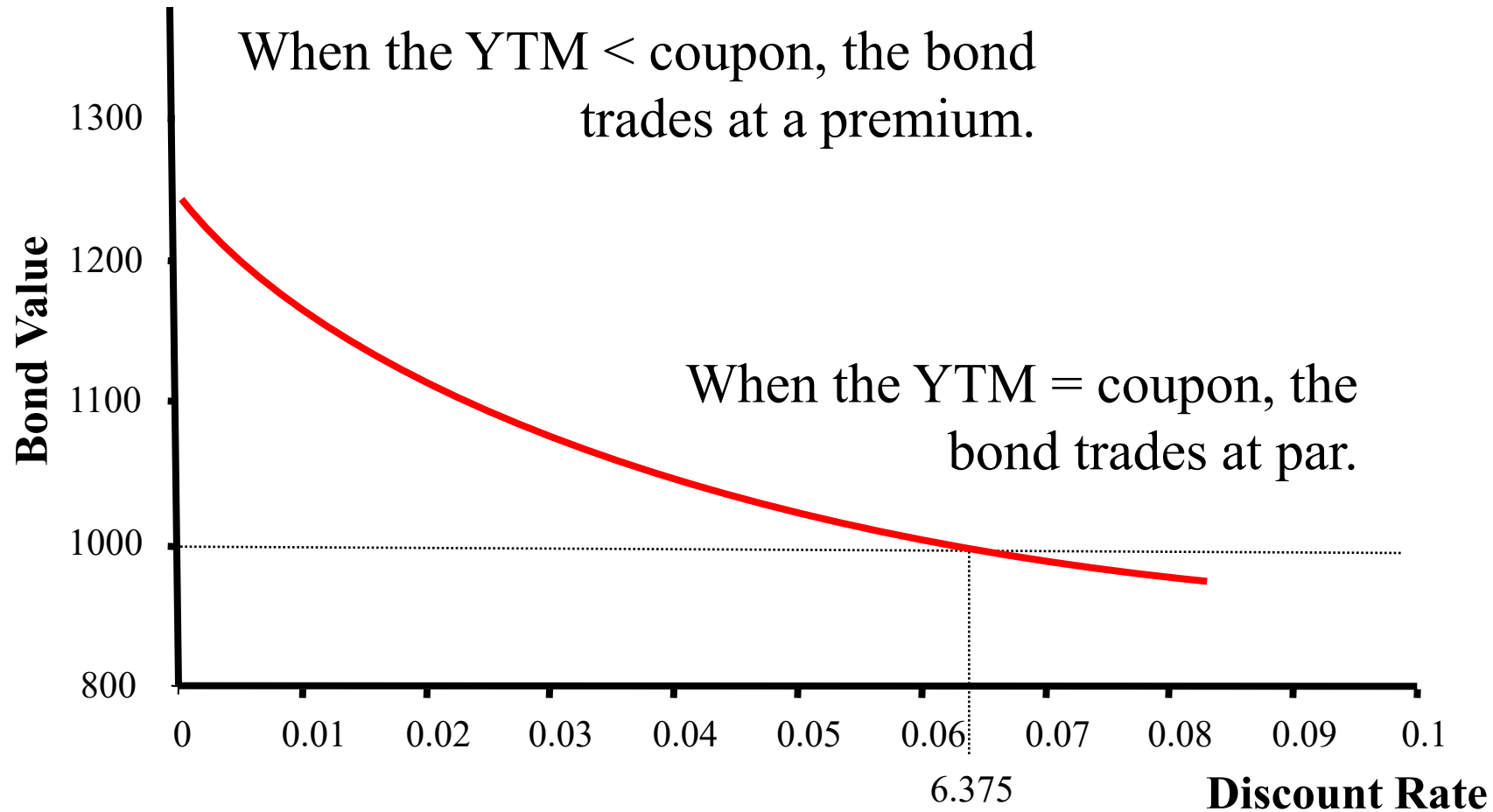
Example 3:

- Now assume that the required yield is 11%.
- How does this change the bond's price?



$$PV = \frac{\$31.875}{.11/2} \left[1 - \frac{1}{(1.055)^{10}} \right] + \frac{\$1,000}{(1.055)^{10}} = \$825.69$$

YTM and Bond Value



When the YTM > coupon, the bond trades at a discount.

Bond Concepts

- ❑ Bond prices and market interest rates move in opposite directions.
- ❑ When coupon rate = YTM, price = par value
- ❑ When coupon rate $>$ YTM, price $>$ par value (premium bond)
- ❑ When coupon rate $<$ YTM, price $<$ par value (discount bond)

Example 4

- What is the price of a bond if:
 - A) Yield is 6% p.a. compounding semi-annually
 - B) Yield is 5% p.a. compounding semi-annually
 - C) Yield is 4% p.a. compounding semi-annually
- The bond has a face value of \$100,000 and has five years until maturity. It has a coupon rate of 5% p.a. payable semi-annually?
- Step 1 Calculate the coupon payments and term
 - Coupon Pmt's = $\$100,000 \times 5\% \div 2 = \$2,500$
 - Term = $5 \times 2 = 10$

Example 4 Solution

❖ $n=10$; $i=?$; $FV=100,000$; $PMT=2,500$

▶ A) Price at 6% = \$95,734.90

□ If coupon rate < Yield then Price < Face value

□ Discount Bond

□ B) Price at 5% = \$100,000.00

□ If coupon rate = Yield then Price = Face Value

□ Par Value Bond

□ C) Price at 4% = \$104,491.29

□ If coupon rate > Yield then Price > Face Value

□ Premium Bond

Computing Yield to Maturity

- Yield to maturity is the rate implied by the current bond price.
- Finding the YTM requires trial and error if you do not have a financial calculator and is similar to the process for finding r with an annuity.
- If you have a financial calculator, enter N , PV , PMT , and FV , remembering the sign convention (PMT and FV need to have the same sign, PV the opposite sign).

YTM with Annual Coupons

- Consider a bond with a 10% annual coupon rate, 15 years to maturity, and a par value of \$1,000. The current price is \$928.09.
 - Will the yield be more or less than 10%?

YTM with Annual Coupons

$$\begin{aligned} \text{PV} &= 100 \left[\frac{1 - \frac{1}{(1+r)^{15}}}{r} \right] + \frac{1000}{(1+r)^{15}} \\ &= 928.09 \end{aligned}$$

$$\text{YTM}=r= 11\%$$

YTM with Semiannual Coupons

- Suppose a bond with a 10% coupon rate and semiannual coupons has a face value of \$1,000, 20 years to maturity, and is selling for \$1,197.93.
 - Is the YTM more or less than 10%?
 - What is the semi-annual coupon payment?
 - How many periods are there?

YTM with Semiannual Coupons

$$\begin{aligned} PV &= 50 \left[\frac{1 - \frac{1}{(1 + r)^{40}}}{r} \right] + \frac{1000}{(1 + r)^{40}} \\ &= 1197.93 \end{aligned}$$

$$r = 4\%$$

$$\text{YTM} = 2r = 8\%$$

3. **Bond Yields** Skolits Corp. issued 15-year bonds two years ago at a coupon rate of 5.1 percent. The bonds make semiannual payments. If these bonds currently sell for 105 percent of par value, what is the YTM?

Here we are finding the YTM of a semiannual coupon bond. The bond price equation is:

$$P = \$1,050 = \$25.50(\text{PVIFA}_{R\%,26}) + \$1,000(\text{PVIF}_{R\%,26})$$

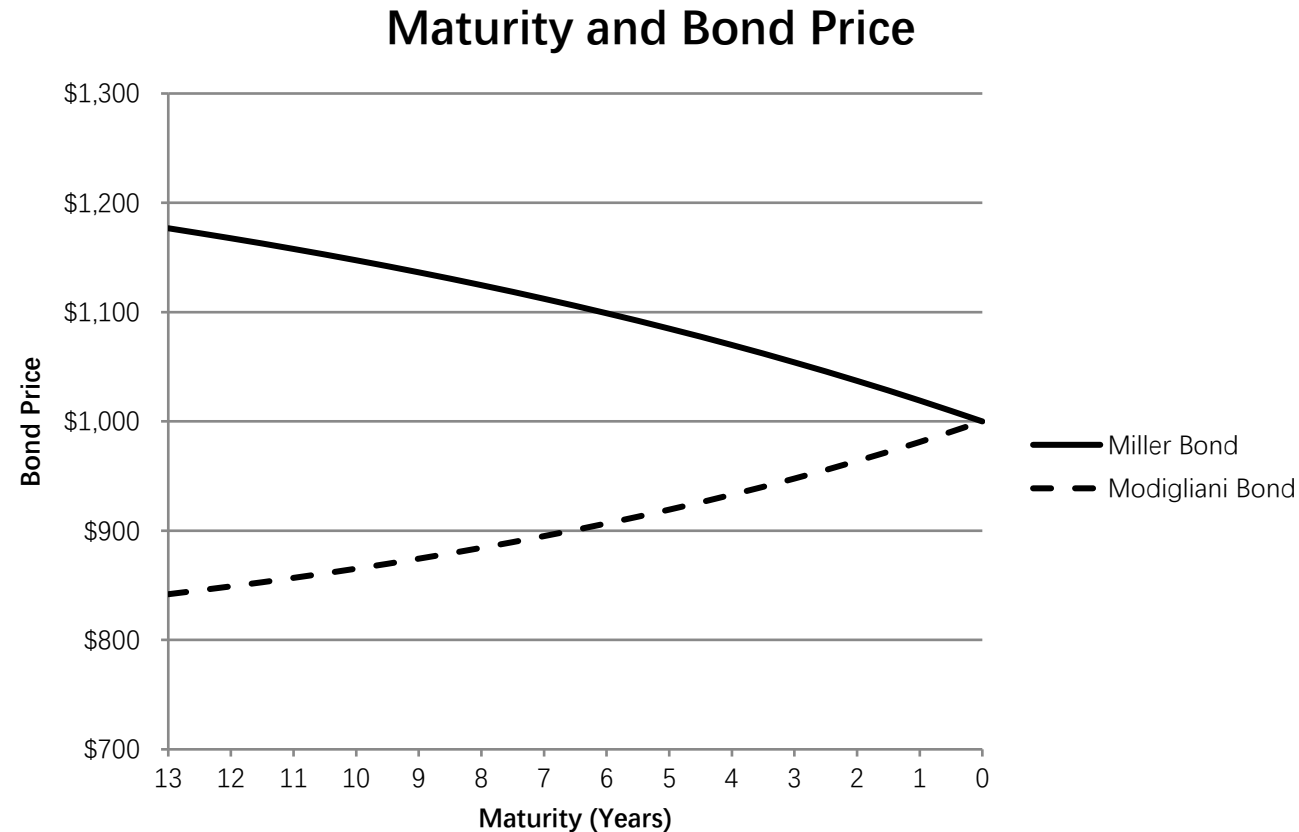
× Since we cannot solve the equation directly for R , using a spreadsheet, a financial calculator, or trial and error, we find:

$$R = 2.293\%$$

Since the coupon payments are semiannual, this is the semiannual interest rate. The YTM is the APR of the bond, so:

$$\text{YTM} = 2 * 2.293\% = 4.59\%$$

16. Bond Price Movements Miller Corporation has a premium bond making semiannual payments. The bond has a coupon rate of 8.2 percent, a YTM of 6.2 percent, and 13 years to maturity. The Modigliani Company has a discount bond making semiannual payments. This bond has a coupon rate of 6.2 percent, a YTM of 8.2 percent, and also has 13 years to maturity. If interest rates remain unchanged, what do you expect the price of these bonds to be 1 year from now assuming both bonds have a par value of \$1,000? In 3 years? In 8 years? In 12 years? In 13 years? What's going on here? Illustrate your answers by graphing bond prices versus time to maturity.



19. Bond Yields Bonino Software has 6.4 percent coupon bonds on the market with 11 years to maturity. The bonds make semiannual payments and currently sell for 108 percent of par. What is the current yield on the bonds? The YTM? The effective annual yield?

Current yield = Annual coupon payment/Price

Current yield = \$64/\$1,080

Current yield = .0593, or 5.93%

The bond price equation for this bond is:

$$P_0 = \$1,080 = \$32(\text{PVIFA}_{R\%,22}) + \$1,000(\text{PVIF}_{R\%,22})$$

Using a spreadsheet, financial calculator, or trial and error we find:

$$R = 2.712\%$$

This is the semiannual interest rate, so the YTM is:

$$\text{YTM} = 2 \times 2.712\%$$

$$\text{YTM} = 5.42\%$$

$$\text{Effective annual yield} = (1 + .02712)^2 - 1$$

$$\text{Effective annual yield} = .0550, \text{ or } 5.50\%$$

28. Valuing Bonds The Grimm Corporation has two different bonds currently outstanding. Bond M has a face value of \$20,000 and matures in 20 years. The bond makes no payments for the first six years, then pays \$800 every six months over the subsequent eight years, and finally pays \$1,000 every six months over the last six years. Bond N also has a face value of \$20,000 and a maturity of 20 years; it makes no coupon payments over the life of the bond. If the required return on both these bonds is 5.9 percent compounded semiannually, what is the current price of Bond M? Of Bond N?

$$\begin{aligned} P_M &= \$800(\text{PVIFA}_{2.95\%,16})(\text{PVIF}_{2.95\%,12}) + \\ &\$1,000(\text{PVIFA}_{2.95\%,12})(\text{PVIF}_{2.95\%,28}) + \$20,000(\text{PVIF}_{2.95\%,40}) \\ P_M &= \$17,791.23 \end{aligned}$$

$$\begin{aligned} P_N &= \$20,000(\text{PVIF}_{2.95\%,40}) \\ P_N &= \$6,251.38 \end{aligned}$$